Dolby Vision Profiles and Levels

Specification

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Notices

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1

Introduction to Dolby Vision bitstream profiles and levels

This document defines Dolby Vision bitstream profiles and levels. Dolby Vision profiles and levels are designed to facilitate implementation of a Dolby Vision product, such as an encoder or decoder, based on consideration of various requirements from typical multimedia applications.

The Dolby Vision profiles provide a rich feature set to support various ecosystems, such as over-the-top streaming, Blu-ray Disc, and 3D-stereoscopic display. Dolby Vision deliverables based on these profiles support many different device Implementation types, such as graphics processing unit (GPU) accelerated software Implementations, full-fledged hardware Implementations, and hardware/software combinations. Dolby Vision profiles support progressive video only. Interlaced video is not supported, and in particular, interlaced coding features must not be used. It is not necessary to implement an application capable of supporting the complete Dolby Vision feature set. By way of example, a display device that does not support 3D-stereoscopic nor cinema-sourced content, need not support profile 20. A limited number of subsets of Dolby Vision features are stipulated by means of bitstream profiles and levels. Refer to the appropriate Dolby Vision kit for more information about which profiles and levels are important to support in your product.

A Dolby Vision profile is composed of:

- A video codec profile (such as HEVC Main10).
- A representative Dolby Vision bitstream profile string.
- Dolby Vision composer metadata and Dolby Vision content metadata carried and encoded in a specified fashion appropriate for the codec. Dolby Vision metadata may be carried as a private Network Abstraction Layer (NAL) unit, a standardized and/or private supplemental enhancement information (SEI) message, or other carriage methods appropriate for elementary streams of a particular video codec.
    - To deliver desired user experiences, some Dolby Vision profiles may require Dolby Vision playback devices to support specific Dolby Vision metadata.

Certain Dolby Vision profiles support a cross-compatible base layer based on video elementary stream metadata, such as video usability information (VUI). Such bitstreams:

- Can be played by a decoder system that is unaware of Dolby Vision, using only the base layer
- Result in a standards-based base-layer video signal, such as HDR10, hybrid log-gamma (HLG), or BT.709 standard dynamic range (SDR), using video elementary stream metadata (for example, HEVC VUI and/or standardized and/or private SEI message, or other carriage methods appropriate for elementary streams of a particular video codec)
- Imply additional (potentially duplicate) stream signaling for a base layer and an enhancement layer
- Can be played by a standards-based decoder system, upon removal of each Dolby Vision element

Note: This documentation does not define standards-based dynamic HDR bitstreams such as those specified by ATSC, DVB, ETSI, or as may be specified in the future by other standards.

Separate specifications provide additional information on use of containers with Dolby Vision. In particular, the *Dolby Vision Streams Within the ISO Base Media File Format* specification describes use with ISOBMFF files of standard and Dolby Vision-specific codecs in the sample entry, and standard and Dolby Vision-specific configuration boxes.

This specification allows a content service provider to enable optimizations for devices that support cinema-sourced and/or stereoscopic 3D content using a Dolby Vision profile, profile 20, based on HEVC Multiview Main10. 3D stereoscopic video (in other words, zero degree of freedom 3D video) is supported by delivering separate left and right images using the HEVC Multiview codec. This support

extends to both cinema-graded and home-graded content. This specification also defines Dolby Vision 2D content that has been graded for cinematic presentation.

- Version history
- Standards and Dolby documentation
- Contacting Dolby

1.1 Version history

Use the version history to track updates made to past revisions of the specification.

For v1.5.0, the changes include:

- Added support for use of CCID = 4 with Dolby Vision profile 20.
- Changes in table in Constraints on codec level. For profile 20, adjusted the maximum allowed Dolby Vision level from 12 to 13.
- Deprecated Dolby Vision profile 4.
- Deprecated use of CCID = 2 with Dolby Vision profiles 8 and 10. (Note that Dolby Vision profile 9.2 is still supported).
- Repurposed section 5 to describe low level mapping related to reference processing unit's `vdr_rpu_profile`, and based on the cross-compatibility of the bitstream's base-layer.
- Clarifications on allowed VUI values for profile 10 with CCID = 0.
- Removed references to H.266 Versatile Video Coding (VVC). Dolby anticipates defining a future Dolby Vision VVC profile.
- Other clarifications and errata.

For v1.4.0, the changes include:

- Definition of Dolby Vision profile 20 with support for CCID = 0, and HEVC and HEVC Multiview codecs. Constraints added to allow improved consumer experience with certain cinema-sourced content.
- New columns for Table 1: *Dolby Vision bitstream profiles* to provide higher-level context of metadata carriage mechanism and supported metadata compression for each profile.
- For CCID = 0, alternative VUI values allowed, including `15` for color matrix, representing IPT-PQ-C2.
- For CCID = 4, clarifications related to alternative transfer characteristics and `preferred_transfer_characteristic`.
- Clarifications of semantics for maximum pixel rate in Dolby Vision levels to allow averaging over a one-second period, and highlighting the differences in time frames between Dolby Vision levels and MPEG levels.
- Clarifications related to Dolby Vision levels and MPEG levels in Dolby Vision levels and Dolby Vision level ID. In particular, high-level explanation is provided as to why MPEG AVC, HEVC, and VVC levels are not comparable to Dolby Vision levels.
- Simplified Annex III's references to alphabetic string names.

For v1.3.6, the changes include:

- Clarifications related to the semantics for *BL signal cross-compatibility ID* in section 2.1 Dolby Vision bitstream profiles.
- Errata related to section *2.1.1 Note to profiles* in *Notes to Profile 5*.

1.2 Standards and Dolby documentation

Standards and Dolby documentation provide additional information to assist you in designing your product.

These are the standards relevant to this documentation:

- *AV1 Specification*, v1.0.0, 2018-06-25 from the Alliance for Open Media, available from https://github.com/AOMediaCodec/av1-spec.
- CTA-861-I (2023), *A DTV Profile for Uncompressed High Speed Digital Interfaces*, available from https://www.cta.tech.
- ETSI GS CCM 001 v1.1.1 (2017-02), *Compound Content Management Specification*, available from https://www.etsi.org/standards.
- ETSI TS 102 366, v1.4.1 (2017-09), *Digital Audio Compression (AC-3, Enhanced AC-3) Standard*; *Annex H - Extensible format for delivery of metadata*; available from http://www.etsi.org/standards.
- EBU R 103, version 3.0, May 2020, *Video Signal Tolerance in Digital Television Systems*, available from https://tech.ebu.ch/home.

- ITU-R BT.709, *Parameter Values for the HDTV Standards for Production and International Program Exchange*, available from https://www.itu.int
- ITU-R Recommendation BT.2020, *Parameter Values for Ultra-High Definition Television Systems for Production and International Program Exchange*, available from https://www.itu.int.
- ITU-R Recommendation BT.2100-2, *Image Parameter Values for High Dynamic Range Television for Use in Production and International Program Exchange*, available from https://www.itu.int.
- Report ITU-R BT.2390, *High Dynamic Range Television for Production and International Program Exchange*, available from https://www.itu.int.
- ITU-T H.265, *Infrastructure of Audiovisual Services—Coding of Moving Video, High efficiency video coding*, available from http://www.itu.int
- ITU-T H.266 (08/2020), *Infrastructure of Audiovisual Services - Coding of Moving Video, Versatile video coding*, available from http://www.itu.int.
- ITU-T H.274 (08/2020), *Infrastructure of Audiovisual Services - Coding of Moving Video, Versatile supplemental enhancement information messages for coded video bitstreams*, available from http://www.itu.int.
- ISO/IEC 14496-12:2022, *Information Technology—Coding of Audio-Visual Objects, Part 12: ISO Base Media File Format*, available from https://www.iso.org/home.html. This documentation is Part 12 of the MPEG-4 specification and describes storage of content in a media file.
- RFC 6381, *The 'Codecs' and 'Profiles' Parameters for "Bucket" Media Types*, August 2011, available from http://tools.ietf.org/html.
- SCTE 215-1-1 2024, *HEVC Video Constraints for Cable Television, Part 1-1 HDR10 Coding*.
- SMPTE RP-431-2:2011, *D-Cinema Quality—Reference Projector and Environment*, available from http://www.smpte.org.
- SMPTE EG 432-1:2010, *Digital Source Processing—Color Processing for D-Cinema*, available from http://www.smpte.org.
- SMPTE ST 2084:2014, *High Dynamic Range Electro-Optical Transfer Function of Mastering Reference Displays*, available from http://www.smpte.org.
- SMPTE ST 2086:2018, *Mastering Display Color Volume Metadata Supporting High Luminance and Wide Color Gamut Images*, available from http://www.smpte.org.
- SMPTE ST 2128:2023, *Colorimetry of IPT-PQ-C2 Color Space for High Dynamic Range and Wide Color Gamut Images*, available from http://www.smpte.org.
- *4cc codes* as registered at http://mp4ra.org/#/codecs.
- *Dolby Vision Streams Within the ISO Base Media File Format*
- *Dolby Vision Streams Within the MPEG-2 Transport Stream Format*
- *Dolby Vision Streams within the MPEG-DASH Format*
- *Dolby Vision streams within the HTTP Live Streaming Format*
- *Dolby Vision Metadata Levels, May 14, 2021* available from https://professionalsupport.dolby.com/s/article/Dolby-Vision-Metadata-Levels.

1.3 Contacting Dolby

Support services are available to address any questions and to provide advice about integrating Dolby technology into your product.

For product design or testing, contact Dolby at systemsupport@dolby.com. By utilizing Dolby expertise, especially during the design process, many problems that might require design revisions before a product is approved can be prevented.

Dolby is also available to review product plans, including preliminary design information, markings, displays, and control and menu layouts, with the goal of preventing problems early in the product development cycle.

If you have comments or feedback about this documentation, send us an email at documentation@dolby.com.

2

Dolby Vision profiles and levels

Dolby Vision profiles and levels are defined by Dolby to specify possible feature configurations for a Dolby Vision stream.

- Dolby Vision bitstream profiles
- Dolby Vision levels
- Dolby Vision codec string

2.1 Dolby Vision bitstream profiles

A Dolby Vision profile is a subset of Dolby Vision feature configurations predefined by Dolby.

Note: Read *Notes to profiles* before employing the Dolby Vision bitstream profiles defined below.

Table 1: Dolby Vision bitstream profiles

Dolby Vision bitstream profile ID	Representative Dolby Vision bitstream profile string	BL/EL codec	BL:EL	BL signal cross-compatibility ID (CCID for Pro Tools and content creation)	Metadata carriage mechanism	Metadata compression supported
5	`dvhe.05`	10-bit HEVC	N/A	0	Unspecified NALu	None
7	`dvhe.07`	10-bit HEVC	1:¼ for UHD; 1:1 for FHD	6	Unspecified NALu	None
8	`dvhe.08`	10-bit HEVC	N/A	1 or 4	Unspecified NALu	None or limited [a]
9	`dvav.09`	8-bit AVC (High or High Progressive profile)	N/A	2	Unspecified NALu	None or limited
10	`dav1.10`	10-bit AV1	N/A	0, 1 or 4	ITU-T T.35 metadata OBU with an EMDF wrapper	None or limited [a]
20	`dvh1.20`	10-bit MV-HEVC (for 3D) or HEVC (for 2D)	N/A	0 or 4	Unspecified NALu	None

[a] Limited metadata compression for Dolby Vision profiles 8 and 10 may not be supported on all playback platforms.

The columns in this table include:

- **Representative Dolby Vision bitstream profile string**: Contains information about the associated profile. For single-layer profiles, this represents the codec of the base layer. For dual-layer profiles, this represents the codec of the enhancement layer (irrespective of whether the profile has cross-compatibility). The Dolby Vision bitstream profile string does not represent a description of a standards-based codec. For AVC and HEVC, these codec profile strings represent unspecified NAL units type as allowed with AVC by ISO/IEC 14496-15:2017, Fourth Edition, 2017-02-01; Amendment 1, 2018-02, section 5.2, 6.2, and Annex F, and with HEVC by ISO/IEC 23008-2:2017, section 7.4.2.2. They have been registered with the MP4 registration authority. For details, see *Dolby Vision profile string* and *Dolby Vision Streams Within the ISO Base Media File Format*.

  Alphabetic versions of profile strings that historically were used for asset management and file names can be found in *Annex III*.
- **BL/EL codec**:
  - 8-bit AVC: H.264 High profile, High Progressive profile, or Constrained High profile
  - 10-bit HEVC: H.265 Main10 profile or H.265 Multiview Main10 profile
  - 10-bit AV1: Based on compatibility with AV1 main profile (10-bit)

  Note: As Dolby Vision supports progressive video only, encoding tools must not use codec features that rely on field or interlaced coding.
- **BL:EL**: Indicates the resolution ratio of base layer to enhancement layer. When N/A, this profile has no enhancement layer.

- **Dolby Vision bitstream profile ID**: A decimal representation of a Dolby Vision profile. Profile IDs 11 to 19 and 21 to 99 are reserved.
- **BL signal cross-compatibility ID (CCID for Pro Tools and content creation)**: For content creation and Pro Tools, an identification number that can be used as a shorthand for a particular form of a base-layer substream that can be decoded to a signal compliant with a particular set of standards, if any. The cross-compatibility ID (CCID) may be used to identify assets to encoders during content creation, or to aid in packaging bitstreams during content distribution (for example, ISO base media file format with MPEG-DASH, or fragmented MP4 with HLS) Dolby Vision encoders must use only the baseline profile composer for incompatible profiles. The behavior of a Dolby Vision product may not always be dependent on these Dolby Vision defined cross-compatibility IDs. In some cases the cross-compatibility IDs are used during content distribution to define packaging elements, such as ISO base media file format `codec_type` boxes. The base layer signal cross-compatibility ID mapping to standards is listed as follows:
  - 0

    None, Dolby Vision proprietary 10 bit.

    For the color space, this uses BT.2020-2. For the transfer characteristic, 'PQ with reshaping' is used. For the color matrix, IPT-PQ-C2 is used, as defined in SMPTE ST 2128:2023. It uses 4:2:0 chroma sub-sampling.

    Certain VUI values may use 'unspecified' (`2`) or, optionally, the values listed in *Cross-compatibility ID to VUI mapping* .
  - 1

    CTA HDR10, as specified by EBU TR 038: HDR10, specifies the use of the perceptual quantization electro-optical transfer function (EOTF) (SMPTE ST 2084) with 10-bit quantization, an ITU-R BT.2020 / BT.2100-2 color space, mastering display color volume as specified in SMPTE ST 2086, and optional static metadata parameters maximum frame-average light level/maximum content light level (MaxFALL/MaxCLL). It uses a limited-range video signal. It is referred to as PQ10 when the static metadata are not used, as might be the case for a live application. Additionally, for Dolby Vision systems, P3 color gamut information is sent using the BT.2020 container. Also, it uses YCbCr 4:2:0 sampling.

    We strongly recommend that bitstreams with a cross-compatibility ID of `1` include ST 2086 metadata in an MPEG SEI message to facilitate broader applications of the bitstreams (for example, transmission over ATSC 3.0).

    ITU-R BT.2100 provides an additional specification of the EOTF, color subsampling, and signal range.
  - 2

    SDR: BT.1886, ITU-R BT.709, YCbCr 4:2:0
  - 3

    Reserved for Dolby Vision proprietary, non-SDR and non-HDR base layer.
  - 4
    - For certain broadcast and mobile systems, a transfer characteristic VUI value of `18` may provide base-layer compatibility that works best with certain classes of devices. This uses a BT.2100-2 gamut in ITU-R BT.2020, NCL Y'CbCr 4:2:0, and assumes non-SDR backward-compatible HLG signaling, as defined in H.265, ITU-R BT.2100, ATSC3, and ARIB STD-B67. The recommended chroma sample location type VUI is `2` (top-left).
    - For other broadcasts systems, a transfer characteristic VUI value of `14`, as per ETSI TS 101 154, v2.5.1 (2019-01) and subsequent versions (and optionally 1, 6, or 15) may provide the best base-layer SDR backward-compatible HLG signaling when used with the `alternative_transfer_characteristic` SEI message, at every random access point, with the `preferred_transfer_function` set to `18`, as per ETSI TS 101 154, v2.5.1. The recommended chroma sample location type VUI is `0` (center-left).

      Note: According to section D.3.38 (*Alternative transfer characteristics SEI message semantics*) of the H.265 specification, if a playback device does not support playback processing per the value of `18` signaled in the alternative transfer characteristics SEI message `preferred_transfer_`

`characteristic` value, then, as described here, it may alternatively use the approach described by `14` in the bitstream transfer characteristics VUI.

ITU-R BT.2100 provides an additional specification of the transfer characteristic, color subsampling, and signal range.

- 5

  Reserved.
- 6

  Ultra HD Blu-ray Disc HDR (per Blu-ray Disc Association standard).
- 7:

  Reserved.
- 15:

  Reserved for additional Dolby Vision proprietary, non-SDR and non-HDR base layer.

  Each base layer signal cross-compatibility ID is related to a VUI value. The mapping is listed in the following table.

Table 2: Cross-compatibility ID to VUI mapping

BL signal cross-compatibility ID	Cross-compatibility ID label	Type of cross-compatibility	VUI
0	Dolby Vision proprietary 10-bit	None	For base layer of profile 5, if including VUI, two options are available: - Preferred [a]: `1,9,16` [b],15,`0` [c] - Original: `1,2,2,2,0` [c] For base layer of profile 10, if including VUI, two options are available: - Preferred [a]: `1,9,16` [b],15,`1` [c] - Original: 1,NA [d], NA, NA, `1` [c] For base layer of profile 20: `1, 9, 16` [b], `15` [e], `2` [c]
1	HDR10	HDR10	For base layer of profile 8: `0,9,16,9,0` [c] For base layer of profile 10: `0,9,16,9,1` [c]
2	SDR	SDR	For base layer of profile 9: `0,1,1,1,0` [c]
4	HLG	HLG	For ARIB base layer of profile 8, 10 and 20: `0,9,18,9,2` [c] For DVB BT.2020 base layer of profile 8: `0,9,14,9,0` [c]
6	Blu-ray	Blu-ray	For base layer of profile 7: `0,9,16,9,2` For enhancement layer of profile 7: `0,9,16,9,2`

[a] Preferred as of *Dolby Vision Profiles and Levels, v1.4, 29 September 2023*.

[b] Although `16` for transfer characteristic generally indicates perceptual quantization (PQ), in the context of Dolby Vision `CCID=0` when `color_matrix` is `15`, (which indicates IPT-PQ-C2), then the actual proprietary transfer characteristic, even when signaled with `16`, is 'PQ with reshaping'. IPT-PQ-C2 color matrix is defined in STMPTE ST 2128:2023.

[c] Default value is shown in the last digit for chroma siting location; however, other values are allowed.

[d] Here, NA represents an unspecified VUI value for AV1.

[e] `15` represents IPT-PQ-C2, as defined in STMPTE ST.2128.

For both base layer and enhancement layer, the comma-separated five-part VUI value represents range, color primaries, transfer characteristic, matrix, and chroma sample location type, respectively. Note that AV1 uses the term of 'chroma sample position (CSP)' rather than the 'chroma sample location type' used by AVC and HEVC.

For VUI value definition, see ITU-T H.265. Dolby Vision sometimes uses unspecified VUI values to signal some Dolby Vision specific characteristics. Take profile 5 as an example: the VUI value of `1,2,2,2,0`, as defined in ITU-T H.265, represents full-range, unspecified, unspecified, unspecified, and center-left siting. This specification further defines the unspecified VUI values of profile 5; for more information, see *Notes to profiles*. Note that the AV1 specification does not indicate unspecified in the same way as AVC or HEVC allows use of `2` for unspecified. Therefore, three VUI digits are labeled as NA for AV1 with CCID = `0`.

For certain profiles, VUI parameters are required, as bitstreams employing these profiles have a non-SDR base layer. For other Dolby Vision profiles, VUI parameters are optional. For detailed information, see *Notes to profiles* .

Dolby Vision bitstreams for all profiles other than profile 7 may use center-left or top-left luma-chroma siting (for AV1, `CSP_VERTICAL` or `CSP_COLOCATED`). A chroma sample location type VUI is mandatory unless center-left siting is used. If a chroma sample location type VUI is present, it must be accurate. As of the effective date of this specification, top-left chroma siting is tested during Dolby Vision SoC approval; however, it is not tested during the Dolby Vision device approval process.

Note: H.265 (2018-02) requires top-left chroma siting (VUI = `2`), if the decoded video is intended for interpretation according to ITU-R BT.2020-2 or ITU-R BT.2100-1. Previously, H.265 (2016-12) described the default chroma siting as center left (VUI = `0`).

If the chroma sample location type VUI is used, both fields must be set to the same value, consistent with HEVC requirements for progressive video.

Note: As of the effective date of this specification, all commercially produced profile 5 Dolby Vision bitstreams have used center-left siting during chroma downsampling, and are distributed without the VUI value for chroma sample location type. Those bitstreams are compliant with this specification.

Note: The AV1 codec uses parameters rather than VUI. The values shown here are the equivalent parameter values to each VUI value.

- **Metadata carriage mechanism**: Different Dolby Vision profiles use different mechanisms for carriage of the Dolby Vision reference processing unit. These include:
  - Unspecified NAL units (NALu)
  - T.35 versatile supplementary enhancement information (VSEI) using Dolby terminal provider code.
  - T.35 Metadata open bitstream unit (OBU) using Dolby terminal provider code and wrapping the Dolby Vision reference processing unit in an EMDF
- **Metadata compression**: Dolby has defined two types of metadata compression:
  - Limited metadata compression: A metadata compression scheme that requires minimal buffering in the playback device
  - Extended metadata compression: A metadata compression scheme that uses additional buffering in the playback device, beyond that required for limited metadata compression. Extended metadata compression is not used with any current Dolby Vision profiles, but may be used with a future Dolby Vision profile.

Related information

Notes to profiles on page 12

Annex III: Dolby Vision profiles with alphabetic string names on page 23

2.1.1 Notes to profiles

Certain directives must be taken into consideration when using the Dolby Vision bitstream profiles.

- For profile 5:
  - VUI information is optional for a profile 5 bitstream. As of *Dolby Vision Profiles and Levels, v1.4*, if VUI are present, there are two options that may be used: `1,2,2,2,0` (original) and `1,9,16,15,0`

(preferred). The values of 2 are defined in ITU-T H.265, Annex E Video usability information, where the 2s represent 'unspecified'.
    - Within the Dolby Vision context, a profile 5 bitstream must use perceptual quantization with reshaping for the transfer characteristic, even if setting the transfer characteristic value to 16, uses Dolby Vision previously proprietary IPT color space (now known as IPT-PQ-C2) for color primaries, uses full range for range, and, by default, uses center-left siting for chroma sample location.

      As defined in SMPTE ST 2128:2023, IPT-PQ-C2, previously a Dolby Vision proprietary color space, is similar to BT.2100 ICtCp, where I is similar to I, P similar to Cp, and T similar to Ct. As of the date of this specification, the MPEG Joint Video Experts Team (JVET) working group has standardized the code point for this previously proprietary color space, also known as IPT-PQ-C2, using code point 15.
- For profile 7:
    - Base-layer/enhancement-layer full alignment is required, as documented in the *Blu-ray Disc Association Specifications*.
    - The currently used enhancement layer VUI values are compliant with the *Blu-ray Disc Association's UltraHD Blu-ray Specification*.
    - The specification of top-left chroma siting for the base layer and enhancement layer is compliant with the *Blu-ray Disc Association's UltraHD Blu-ray Specification*.
    - The minimal enhancement layer (MEL) can be used for profile 7 to minimize the processing requirements for the enhancement layer and therefore ensure broader use among UltraHD Blu-ray SoCs. For more information, see *Annex II: Differentiating MEL and non-MEL bitstreams*.
- The Reserved profile is reserved for other video ecosystems and video codecs.
- For profiles 7 and 8.1, VUI parameters are required, as bitstreams employing these profiles have a non-SDR base layer. For other Dolby Vision profiles, VUI parameters are optional. For streams of profiles greater than 10 with a `CCID = 4`, VUI parameters are recommended.
- For profile 8:
    - Certain profile 8.4 bitstreams with `transfer_characteristics = 18` may set the VUI parameter `chroma_loc_info_present_flag` to `false` and do not include the VUI information for chroma sample location type. These bitstreams are compliant. We recommend that:
        - Profile 8.4 bitstreams that employ `transfer_characteristics = 14` include all of the VUI parameters.
        - Profile 8.4 bitstreams that employ `transfer_characteristics = 18` include the VUI parameters for range, color primaries, transfer characteristic, and matrix.
    - Limited metadata compression for profile 8 is currently not supported on some playback platforms.
- For profile 9:

  Dolby Vision profile 9 stream must be compliant to H.264/AVC High profile (described in *A.2.4* in *ITU H.264 specification*). This includes all streams compliant to a subset of H.264 High profile. Streams must be marked in the sequence parameter set as compliant to either of:
    - H.264 High profile
    - H.264 Progressive High profile
    - H.264 Constrained High profile
- For profile 20:
    - Although descriptions of certain elements of profile 20 elementary streams are provided here as an aid in developing a high-level understanding of various forms of profile 20, these are in no way complete descriptions of profile 20 elementary streams. See either the *Dolby Vision Consumer Decoder Specification* or *Dolby Vision Encoder System Implementation Development Kit* documentation for complete descriptions of the elementary streams.
    - Profile 20 playback devices must support certain newer Dolby Vision metadata levels that are not mandatory for profile 5, 8 and 10 playback devices to support. In particular, profile 20 playback devices must support consumption of display management version 4 metadata, including L8, Target Trim Metadata.
    - Certain constraints are used with the Dolby Vision reference processing unit and Dolby Vision metadata levels to provide an improved user experience with 3D-stereoscopic and/or cinema-sourced

video content. See either the *Dolby Vision Consumer Decoder Specification* or *Dolby Vision Encoder System Implementation Development Kit* documentation.

- The Dolby Vision level applies to each layer of a 3D bitstream, and to layer 0 of a 2D bitstream.
- For 3D-stereoscopic elementary streams:
  - `nuh_layer_id` must be set equal to `0` or `1`.
  - `general_profile_idc` must be set equal to `2` or `6`, with the H.265 base layer set equal to `2` and with the H.265 non-base layer set equal to `6`.
  - Other than the eye to which a layer is assigned and the setting for `general_profile_idc`, an identical set of Dolby Vision and H.265 characteristics should be applied to all H.265 layers, including the H.265 base layer.
  - A `three_dimensional_reference_displays_info` SEI message must be present in order to associate layers with left or right eye. This SEI message must be contained in the HEVC decoder configuration record (`'hvcC'`).
    - A Dolby Vision profile 20 player must be tolerant of a bitstream with `num_ref_displays_minus1` set equal to `0`, `1`, `2`, or `3`. A Dolby Vision profile 20 player receiving a Dolby Vision profile 20 bitstream with up to four reference displays must successfully playback as Dolby Vision view `id[0]`, even when additional reference display view pairs are present.

      Note: Though this specification anticipates Dolby Vision profile 20 bitstreams with more than a single left/right view pair, at the time of release of this specification, Dolby Vision encoding tools support a single pair only.
- For 2D elementary streams:
  - `nuh_layer_id` must be set equal to `0`.
  - `general_profile_idc` must be set equal to `2`.
  - Although this specification describes use of Dolby Vision profile 20 bitstreams with 2D video sequences (that is, with `nuh_layer_id` constrained to `0`), approval of Dolby Vision profile 20 playback devices does not require, at this point, successful playback of 2D Dolby Vision profile 20 bitstreams.
  - VUI parameters are required, as bitstreams that employ this profile may have a base layer other than SDR.
  - For `CCID = 0`:
    - Signaled VUI `transfer_characteristic = 16` represents PQ although the actual proprietary transfer characteristic of the base layer is ‘PQ with reshaping’.
    - For color primaries, the stream must use IPTPQ-C2, as defined in SMPTE ST 2128:2023, where I is similar to I, P similar to Cp, and T similar to Ct. A VUI value of `15` is intended to represent IPTPQ-C2. Earlier, this was a Dolby Vision proprietary color space.
  - Dolby Vision profile 20.0 playback devices must be tolerant of the VUI `matrix_coeffs` set either to `2` or `15`.
  - Although Dolby Vision profile 20 playback devices need not support limited or extended metadata compression, they must parse the `dv_md_compression` field of an ISO base media file format stream in the Dolby Vision configuration record. See *Dolby Vision Streams Within the ISO Base Media File Format*, v2.5 or later. If `dv_md_compression` is not equal to `0`, and if the Dolby Vision profile 20.0 playback device does not support playback of bitstreams with compressed metadata, then the Dolby Vision profile 20.0 playback device:
    - Must not present a decoded bitstream
    - Must provide a message to the user that an unrecognizable bitstream has been encountered
- For information about profile 0, 1, 2, 3, 4, and 6, see *Annex I: Profiles not supported for new applications*.

Related information

Annex I: Profiles not supported for new applications on page 23
Annex II: Differentiating MEL and non-MEL bitstreams on page 23
Dolby Vision bitstream profiles on page 9

2.1.2 Dolby Vision profile strings

To signal the profile information of a Dolby Vision bitstream, a Dolby Vision bitstream profile string is used. This profile string follows a predefined naming convention.

A Dolby Vision bitstream profile string is composed in the following pattern:

```
[Codec_type].[bitstream_profile_ID]
```

Table 3: Dolby Vision profile string

Attribute	Value	Description
Codec_type	dvhe, hev1	• dvh* represents the HEVC-based Dolby Vision codecs. • he** and hv** represent the HEVC and MV-HEVC codec.
	dvh1, hvc1	
	dvav, avc3	• dva* represents the AVC-based Dolby Vision codecs. • avc* represents the AVC codec.
	dva1, avc1	
bitstream_profile_ID	05, 07, 08, 09, 20	A representation of the bitstream profile ID.

Dolby Vision profile strings that begin with d use Dolby Vision codecs as defined here. Compatible bitstreams employ a standard FourCC codec string that starts with an h, a, or v. As defined in *Dolby Vision Streams Within the ISO Base Media File Format*, Dolby Vision specific configuration boxes may be used with standard codecs strings for certain profiles. he, av, and vv represent standard codecs as defined at https://mp4ra.github.io/atoms.html, and consistent with ISO/IEC 14496-12:2012. Codecs other than HEVC, AVC, or AV1 may be supported in the future, for which additional Dolby Vision bitstream profile IDs will be added.

Refer to the bitstream profile name column in the *Dolby Vision bitstream profiles* table for examples.

For transmission of Dolby Vision streams within the MPEG-DASH or HLS format, see these specifications: *Dolby Vision Streams Within the MPEG-DASH Format* and *Dolby Vision Streams Within the HTTP Live Streaming Format*.

For transmission of a Dolby Vision stream using Common Media Application Format (CMAF) packaging, and specification of "codec" strings (also known as @codec) for that environment, see *SCTE 215-1-1 2020*.

For certain asset management and production applications, alphabetic versions of Dolby Vision profile strings are used. For more information, see *Annex III*.

Related information

Annex III: Dolby Vision profiles with alphabetic string names on page 23

2.2 Dolby Vision levels

A Dolby Vision level specifies the maximum pixel rate, maximum decoded bitstream video width, and maximum bit rate supported by a product within a given bitstream profile.

Typically, there is a limit on the maximum number of pixels a product can process per second within a given bitstream profile; the Dolby Vision levels defined here generally correspond to the product processing capability. Although not listed, noninteger frame rates are supported.

When considering the following table, note that while values of the maximum pixel rate and the maximum decoded bitstream video width (pixels) columns monotonically increase with each ascending Dolby Vision level ID, the frame rates in the example decoded bitstream resolution at frame rate column do not consistently increase in the same manner.

Table 4: Dolby Vision levels

Level ID	Maximum pixel rate (pps)	Maximum decoded bitstream video width (pixels)	Example decoded bitstream resolution at frame rate (fps)	Maximum Main tier bit rates (Mbps) [a]	Maximum High tier bit rates (Mbps) [a]
01	22,118,400	1280	1280 × 720 @ 24	20	50
02	27,648,000	1280	1280 × 720 @ 30	20	50
03	49,766,400	1920	1920 × 1080 @ 24	20	70
04	62,208,000	2560	1920 × 1080 @ 30	20	70
05	124,416,000	3840	1920 × 1080 @ 60	20	70
06	199,065,600	3840	3840 × 2160 @ 24	25	130
07	248,832,000	3840	3840 × 2160 @ 30	25	130
08	398,131,200	3840	3840 × 2160 @ 48	40	130
09	497,664,000	3840	3840 × 2160 @ 60	40	130
10	995,328,000	3840	3840 × 2160 @ 120	60	240
11	995,328,000	7680	7680 × 4320 @ 30	60	240
12	1,990,656,000	7680	7680 × 4320 @ 60	120	480
13	3,981,312,000	7680	7680 × 4320 @ 120	240	800

[a] The AV1 specification refers to Main and High, rather than Main tier and High tier used in other specifications.

The columns in this table include:

- **Maximum pixel rate (pps)**: This column lists imposed limits on arithmetic combinations of decoded bitstream resolution and frame rate (decoded bitstream resolution multiplied by frame rate: horizontal pixels × vertical pixels × frame rate). The maximum pixels per second is a constant for a given level. The decoded bitstream resolution is inversely proportional to the frame rate, meaning that the decoded bitstream resolution can be reduced for obtaining higher frame rate (and vice versa). Note that the decoded bitstream resolution here is for baseband video, irrespective of the particular video compression codec that is used.

  Note: The maximum pixel rate may be averaged over any one-second period as long as the bitstream instantaneous peak rate, between any two frames, does not exceed the requirements of the bitstream codec levels, as defined in MPEG or other codec specifications. For example, a profile 8.4 bitstream would be compliant even if it signaled Dolby Vision level 9, HEVC level 5.1, and contained 50 frames at 63 fps.
- **Maximum decoded bitstream video width (pixels)**: This column indicates the maximum decoded bitstream video width. This parameter is unique to a Dolby Vision level; it is not a parameter typically specified for codec levels, such as AVC, or HEVC. This parameter is specified for a Dolby Vision level due to constraints that exist in certain Dolby Vision IP cores.
- **Example decoded bitstream resolution at frame rate (fps)**: Base-band picture horizontal and vertical pixels followed by frame rate. Note that the frame rate does not increase monotonically, although the maximum pixel rate does.
- **Maximum bit rates**: This column indicates the maximum combined bit rate of the base and enhancement layers, when applicable.
- **High tier**: Note that for Dolby Vision bitstream profile 7, Blu-ray Disc Association specifications allow a maximum high-tier bit rate of 100 Mbps for each level. Similarly, there may be other Dolby Vision enabled systems that limit or require different maximum bit rates. Additionally, high tier may be required for some applications that use temporal subscale layers.

2.2.1 Dolby Vision level ID

To signal the level information of a Dolby Vision bitstream, the Dolby Vision level ID is used.

Refer to the level ID column in *Dolby Vision levels* for details.

Note: Dolby Vision level IDs are unrelated to MPEG AVC, or HEVC levels and their associated MaxCPB and MaxBR, due to the very distinctive playback device architectures that pertain to a postcompression Dolby Vision processing pipeline and an MPEG reference video elementary stream decoder.

2.3 Dolby Vision codec string

In different use cases, the profile strings and level IDs are presented in different formats for signaling Dolby Vision specific information.

For example, the Dolby Vision codec string is composed in this pattern:

```
[Dolby_Vision_Profile_String].[Dolby_Vision_Level_ID]
```

For detailed information, refer to the *Dolby Vision profile strings* and *Dolby Vision level ID* sections.

Codec string examples:

- `dvav.09.04`

  This string represents a single-layer SDR backward-compatible Dolby Vision stream encoded as 8-bit AVC video with a pixel rate that does not exceed 62,208,000 pixels/second (for example, 1920 × 1080 at 30 fps).
- `dvhe.05.07`

  This string represents a single-layer noncompatible Dolby Vision stream encoded as 10-bit HEVC video with a pixel rate that does not exceed 248,832,000 pixels/second (for example, 3840 × 2160 at 30 fps).
- `dvhe.07.06`

  This string represents a dual-layer Blu-ray HDR10 compatible Dolby Vision stream encoded as 10-bit HEVC video with a pixel rate that does not exceed 299,065,600 pixels/second (for example, 3840 × 2160 at 24 fps).
- `dvhe.08.10`

  This string represents a single-layer backward-compatible Dolby Vision stream encoded as 10-bit HEVC video with a pixel rate that does not exceed 995,328,000 pixels/second (for example, 3840 × 2160 at120 fps).
- `dvh1.20.10`

  This string represents a single-layer noncompatible Dolby Vision stream encoded as 10-bit HEVC or HEVC Multiview video with a pixel rate that does not exceed 995,328,000 pixels per second (for example, 3840 × 2160 at 120 fps). If the stream is intended for 3D-stereoscopic presentation, the ISO base media file format file will include an L-HEVC configuration box (`lhvC`), as documented in *Dolby Vision Streams Within the ISO Base Media File Format*, v2.5 or later.

For detailed information about how to signal Dolby Vision specific information, refer to *Dolby Vision Streams Within the ISO Based Media File Format*, *Dolby Vision Streams Within the MPEG-2 Transport Stream Format*, *Dolby Vision Streams Within the HTTP Live Streaming Format*, and *Dolby Vision Streams Within the MPEG-DASH Format*.

3

Dolby Vision playback device capabilities

Dolby Vision profiles and levels specify typical Dolby Vision stream configurations. A playback device capable of decoding these streams can also advertise its capabilities by using the same Dolby Vision profiles and levels strings.

This table lists example devices and their capabilities specified by Dolby Vision profiles and levels strings.

Example device	Device capabilities
Field-programmable gate array (FPGA)–based TV	- `dvhe.05.07`
First-generation chipset-based TV	- `dvhe.05.07`
First-generation chipset-based ultra-high definition (UHD) Blu-ray player	- `dvhe.07.06` - `dvhe.07.07`
Chipset-based UHD over-the-top (OTT) digital media adapter	- `dvhe.05.09` - `dvhe.08.09`
Chipset-based full-high definition (FHD) set-top box (STB)	- `dvhe.05.07` - `dvhe.08.05`
Chipset-based HD STB	- `dvav.09.04`
Media PCs	- `dvhe.05.09` - `dvhe.08.09`
Chipset-based 8K TV	- `dvhe.05.12` - `dvhe.08.12` - `dvhe.09.05`
Mobile phone that supports only AVC codec at a resolution up to UHD	- `dvav.09.07`
Mobile phone that supports both AVC and HEVC codecs at a variety of resolutions	- `dvhe.05.09` - `dvhe.08.09` - `dvav.09.05`
Chipset-based 3D-stereoscopic display	- `dvh1.20.10`

Every Dolby Vision playback device must pass Dolby Vision system development kit certification. During the certification procedure, the chipset implementing the Dolby Vision decoder will be tested against the advertised device capabilities, and Dolby will approve the device capabilities.

4

Constraints

Certain constraints are imposed by Dolby Vision profiles and levels.

- Constraints on codec level
- Limitation on decoder buffer size

4.1 Constraints on codec level

A Dolby Vision profile can support different level settings. Within a given profile, the maximum level a base layer or enhancement layer can take is restricted by the profile.

The maximum Dolby Vision levels, base-layer codec levels, and enhancement-layer codec levels to which a valid Dolby Vision stream can be set are listed for each Dolby Vision profile.

Table 5: Constraints on codec level

Profile ID	Profile name	BL/EL codec	BL:EL	Dolby Vision level (maximum)	BL/EL codec profile	BL codec level (maximum)	EL codec level (maximum)
5	dvhe.05	10-bit HEVC	NA	13	H.265 Main10	High-tier 6.2	NA
7	dvhe.07	10-bit HEVC	1:1	05	H.265 Main10	High-tier 5.1	High-tier 5.1
			1:¼	09	H.265 Main10	High-tier 5.1	High-tier 5.1
8	dvhe.08	10-bit HEVC	NA	13	H.265 Main10	High-tier 6.2	NA
9	dvav.09	8-bit AVC	NA	05	H.264 High, High Progressive, or Constrained High	High-tier 4.2	NA
20	dvh1.20	10-bit HEVC Multiview	NA	13 [a]	H.265 Main10 Multiview	High-tier 6.2	NA

[a] The Dolby Vision level applies to each layer of a 3D bitstream, and to layer 0 of a 2D bitstream.

Note: Profile 4 was previously listed in this table, but has been deprecated. It is no longer supported for new applications.

In certain cases, the Dolby Vision specification imposes tighter constraints on the maximum tier bit rate and the maximum decoded picture buffer size, as compared to the HEVC Main10 Level 5.1 specification. See sections *Dolby Vision levels* and *Limitation on decoder buffer size*.

Related information

Limitation on decoder buffer size on page 20
Dolby Vision levels on page 15

4.2 Limitation on decoder buffer size

The Dolby Vision levels put limitations on the size of the decoded picture buffer.

In all cases, the maximum number of reference frames is the same for the base layer and enhancement layer.

- For profiles 10 and less, the number of reference frames in the individual layer's decoded picture buffer, whether base layer or enhancement layer, must not exceed six for all levels listed in *Dolby Vision levels*.
- For profiles greater than 10, the maximum decoded picture buffer size must be in accordance with the applicable standard for the codec used for the Dolby Vision profile.

5

Dolby Vision profiles: Low-level definition

The Dolby Vision Consumer Decoder specification provides necessary and contextual information for implementing the low-level Dolby Vision bitstream decoding process. It describes Dolby Vision metadata syntax elements and semantics, and the storage of the metadata in the Dolby Vision reference processing unit (RPU). Also described is the processing of the Dolby Vision base-layer samples.

The `vdr_rpu_profile` parameter is used in the Dolby Vision reference processing unit data header. The following table describes required settings of the `vdr_rpu_profile` parameter based on the Dolby Vision bitstream's base-layer cross-compatibility.

Dolby Vision signal cross-compatibility ID	`vdr_rpu_profile`
0	0 (Baseline)
1, 2, 4 or 6	1

6

Annex

- Annex I: Profiles not supported for new applications
- Annex II: Differentiating MEL and non-MEL bitstreams
- Annex III: Dolby Vision profiles with alphabetic string names

6.1 Annex I: Profiles not supported for new applications

Certain profiles and profile/CCID combinations are not supported for new applications.

These profiles are listed in the following table.

Table 6: Dolby Vision bitstream profiles

Dolby Vision bitstream profile ID	BL signal cross-compatibility ID	Bitstream profile name	BL/EL codec	BL:EL
0	2	`dvav.per`	Advanced Video Coding (AVC)	1:¼
1	0	`dvav.pen`	AVC	1:1
2	2	`dvhe.der`	8-bit HEVC	1:¼
3	0	`dvhe.den`	8-bit HEVC	1:1
4	2	`dvhe.04`	10-bit HEVC	1:¼
6	1	`dvhe.dth`	10-bit HEVC	1:¼
8	5	`dvhe.08`	10-bit HEVC	N/A

For profiles 0 and 1, base-layer/enhancement-layer group-of-pictures alignment is required. For all other dual-layer profiles (profiles 2, 3, 4, and 6), instantaneous-decoder-refresh alignment is required.

For profiles 1, 3, and 6, VUI parameters are required, as bitstreams employing these profiles have a non-SDR base layer. For other Dolby Vision profiles that have an SDR base layer, VUI parameters are optional.

Related information

Notes to profiles on page 12

6.2 Annex II: Differentiating MEL and non-MEL bitstreams

Pictures contained in a Dolby Vision bitstream can be encoded as either MEL or non-MEL, not both. Use the approach described in this section to differentiate the MEL and non-MEL bitstreams.

The MEL consists of Dolby Vision composer and content metadata of a mid-gray flat-field video sequence, carried in a NAL unit.

If a Dolby Vision playback device supports Dolby Vision profile 7, it can check for these values in the reference processing unit of a Dolby Vision bitstream. If the values are not exactly as shown for all three channels, the device can flag the bitstream as a Dolby Vision profile 7 full enhancement layer bitstream; otherwise, flag the bitstream as profile 7 minimum enhancement layer.

Note: Prior to the deprecation of profile 4, this specification also described use of these values by a Dolby Vision playback device that supported profile 4 and chose not to instantiate a second HEVC decoder.

```
rdnp->nlq_offset = 0;
rdnp->vdr_in_max_int = 1;
rdnp->uv.vdr_in_max = 0;
rdnp->up.nlq_linear_dz.linear_deadzone_slope_int = 0;
rdnp->up.nlq_linear_dz.us.linear_deadzone_slope  = 0;
rdnp->up.nlq_linear_dz.linear_deadzone_threshold_int = 0;
rdnp->up.nlq_linear_dz.ut.linear_deadzone_threshold  = 0;
```

Related information

Notes to profiles on page 12

6.3 Annex III: Dolby Vision profiles with alphabetic string names

Previous versions of this specification documented alphabetical string names associated with Dolby profiles. Those specifications noted that they might be appropriate for certain asset management and production use

cases. Those alphabetical string names were never supported in Dolby Vision video elementary streams, Dolby Vision ISO base media file streams, Dolby Vision transport streams, or with Dolby Vision HLS or DASH containers.

We do not endorse the continued use of alphabetical string names, and strongly recommend use of:

- Dolby Vision bitstream profile IDs, as defined in Dolby Vision bitstream profiles
- Dolby Vision base-layer cross-compatibility IDs, as defined in Cross-compatibility ID to VUI mapping
- Dolby Vision MP4 codec values, as defined in the *Dolby Vision Streams Within the ISO Base Media File Format Specification*

Note: It is the responsibility of the user of alphabetical string names to translate to numeric profile names, and if applicable, cross-compatibility IDs (CCIDs) (as defined in *Dolby Vision bitstream profiles*), before interaction with external systems.

Related information

Dolby Vision bitstream profiles on page 9

Glossary

DASH
Dynamic adaptive streaming over HTTP. An adaptive bit-rate streaming protocol that enables high-quality streaming of media content over the Internet delivered from HTTP.

EOTF
Electro-optical transfer function. A generic way of describing a specific function used to convert digital data into light (usually dictated by a particular standard specification). For example, the specification ITU-R BT.1886 describes an EOTF that allows a flat-panel display to simulate the characteristics of a cathode ray tube (CRT) display.

FHD
Full-high definition. Video with a display resolution of 1920×1080 pixels and an aspect ratio of 16:9. Also referred to as 1080p.

HLG
Hybrid log-gamma. High-dynamic range standard format developed jointly by the British Broadcasting Corporation (BBC) and Nippon Hoso Kyokai (Japan Broadcasting Corporation), and defined in ARIB STD-B67 and ETSI TS 101 154.

HLS
HTTP Live Streaming. An adaptive streaming protocol developed by Apple for delivery of media content in various software environments.

MEL
Minimal enhancement layer.

NAL
Network Abstraction Layer.

OTT
Over-the-top. The delivery of audio, video, and other media over the Internet without the involvement of a multichannel video programming distributor (MVPD) or a pay TV operator in the control or distribution of the content.

perceptual quantization
A quantization curve designed to optimize high dynamic range image encoding for maximum efficiency and quality. In this curve, each numeric change falls below the range of human perceptual threshold to avoid perceived image artifacts.

reference processing unit
A container for Dolby Vision metadata that is encoded within a Dolby Vision bitstream. Each frame in a Dolby Vision bitstream contains one reference processing unit.

SEI
Supplemental enhancement information. Data unit that carries supplemental video information about decoding or display, introduced in H.264.

STB
Set-top box.

UHD
Ultra-high definition. Ultra-high-definition television or video, with a display resolution of 3840×2160 pixels in the 16:9 aspect ratio. Also referred to as 2160p.

VUI
Video usability information. A syntax structure that collects information that prepares the decoded video for output and display.

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